

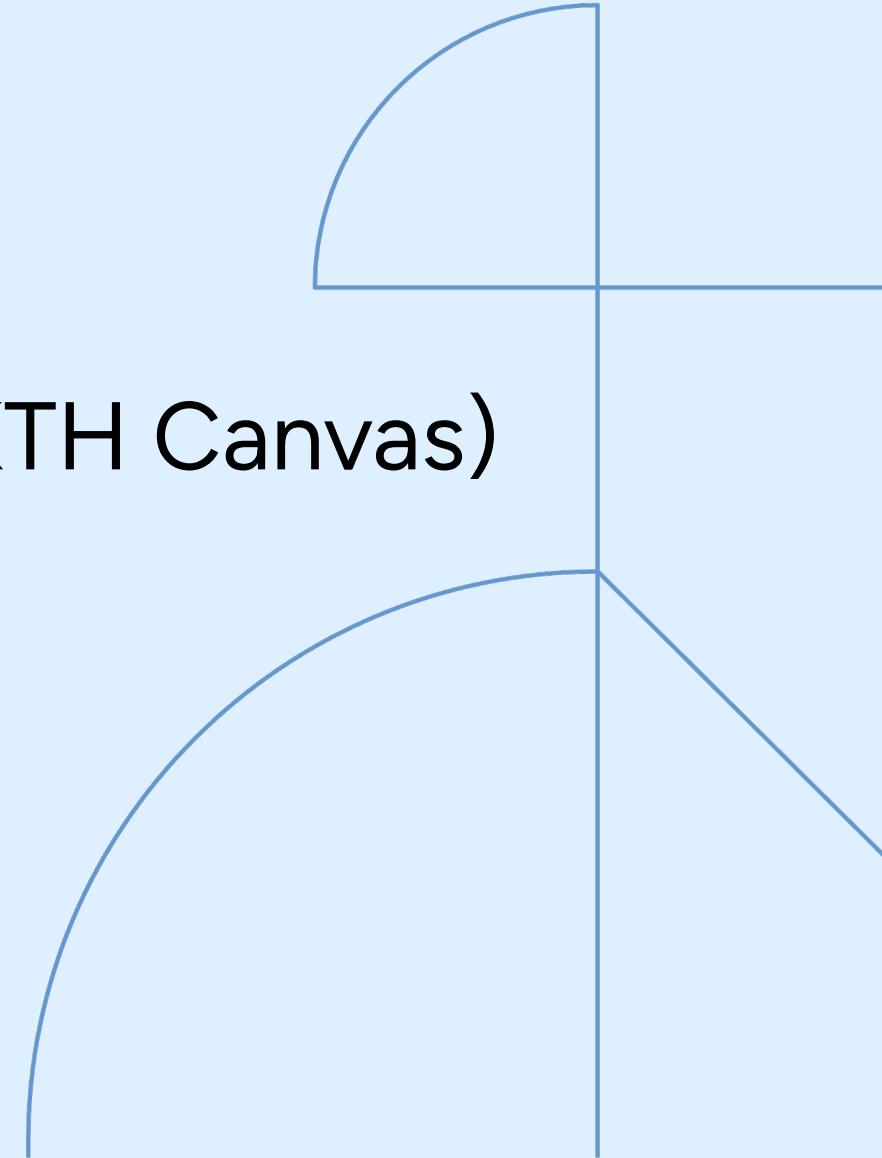
Lab 6
Implementation of a complete IoMT health application

Project Carrier Course, Part 1,

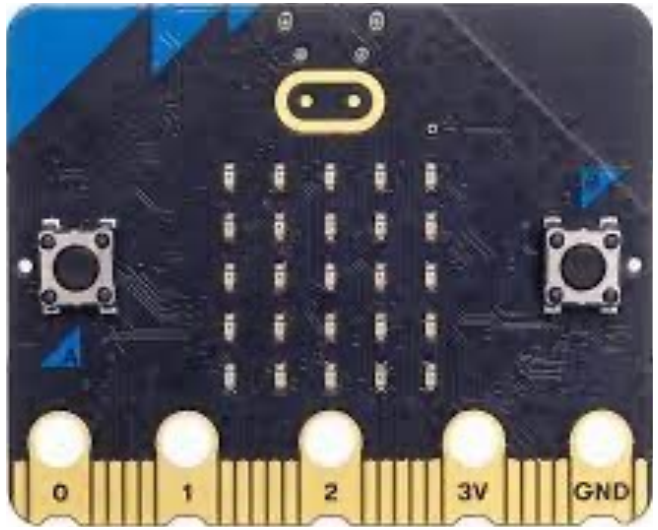
Martin Jacobsson

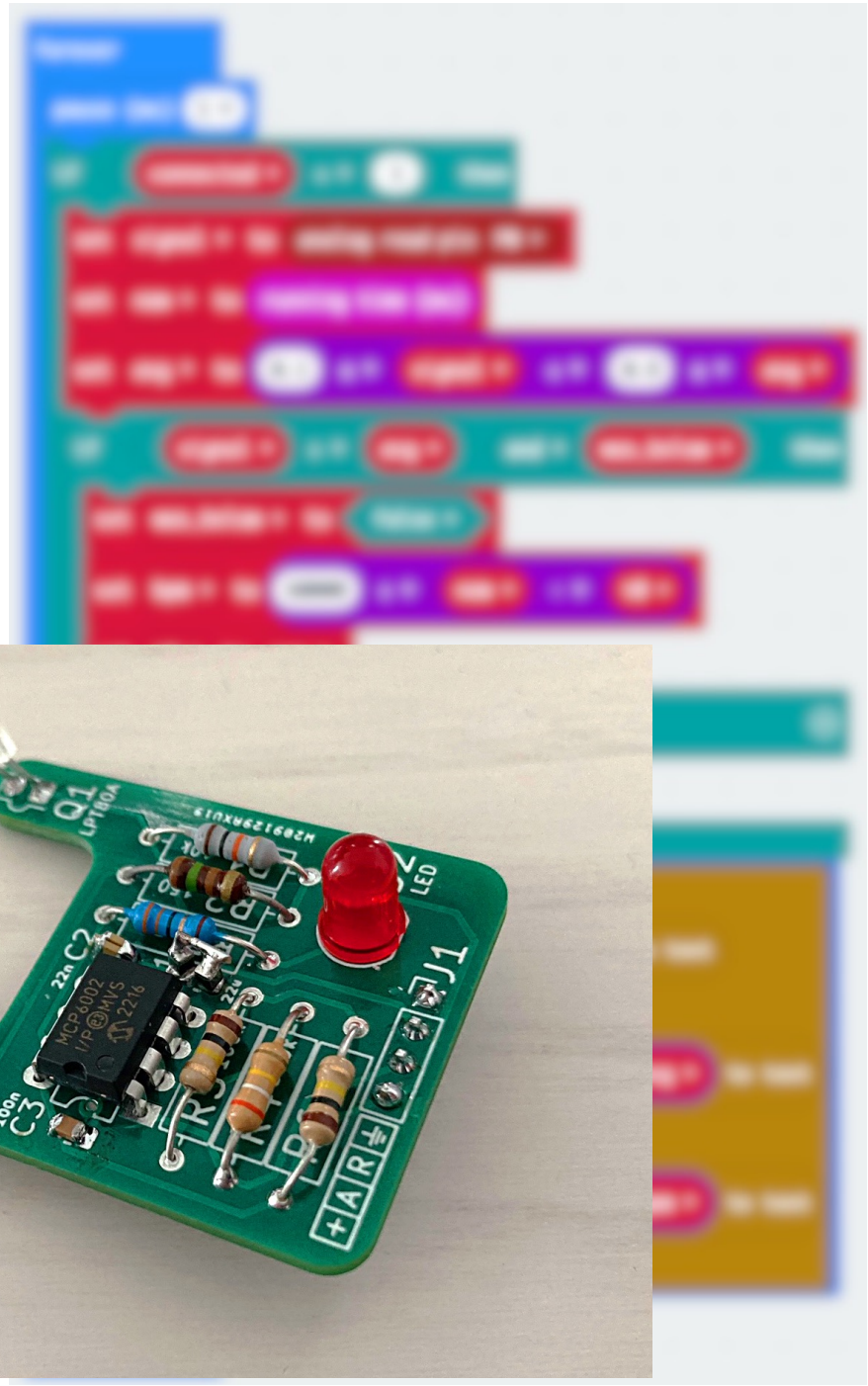
Done in pairs!

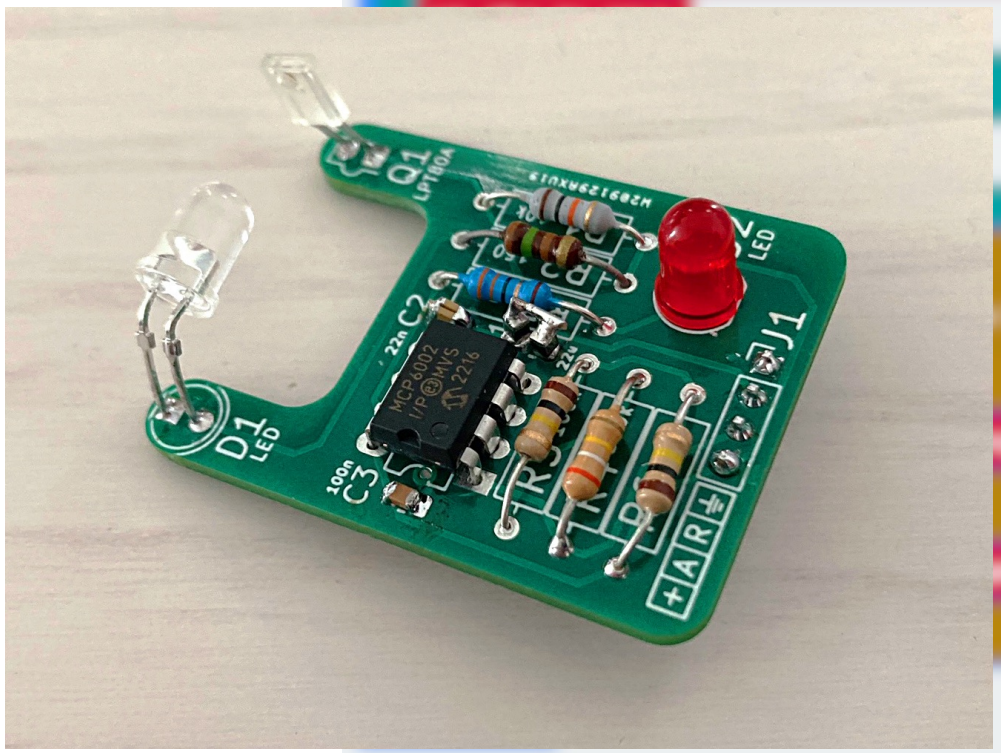
But the new pairs: Pair B (see KTH Canvas)



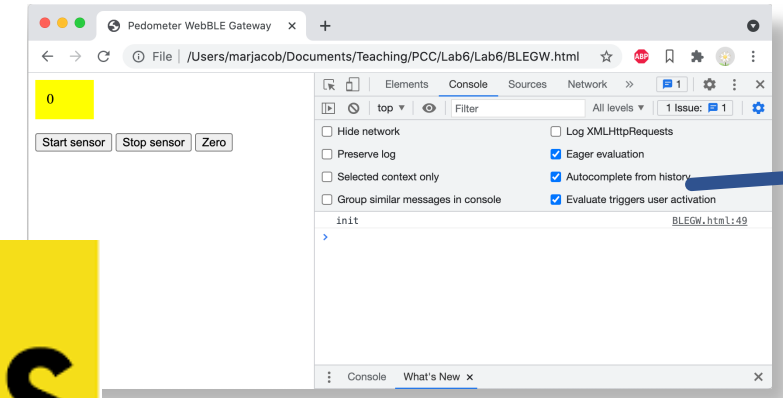
Use Lab 1 + Lab 3 Solution





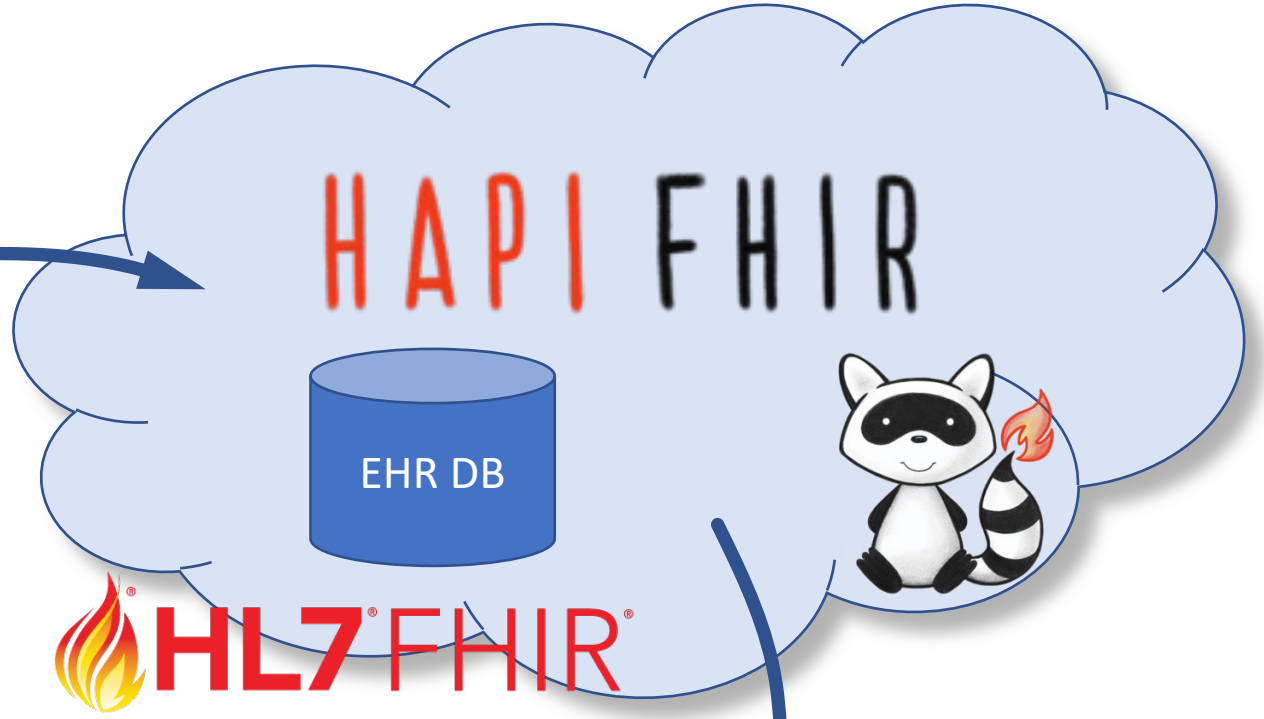


HTML

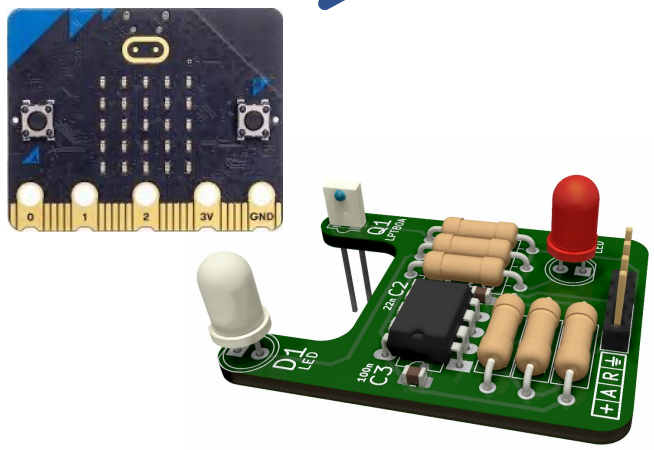


JS

JavaScript +
WebBLE

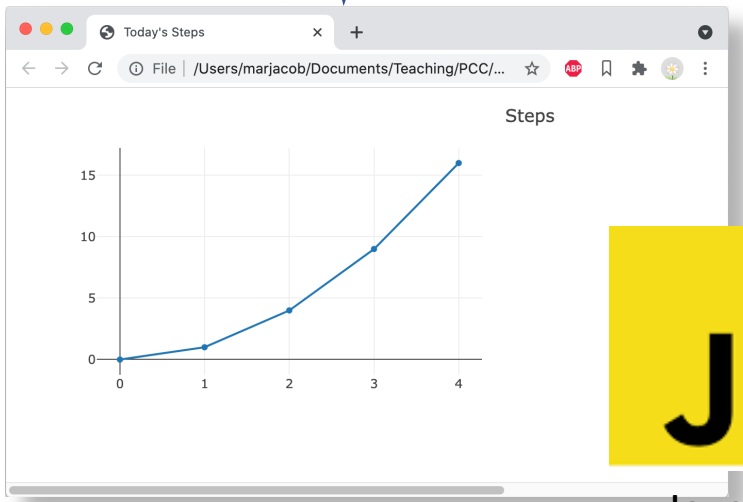


Bluetooth SMART



HTML

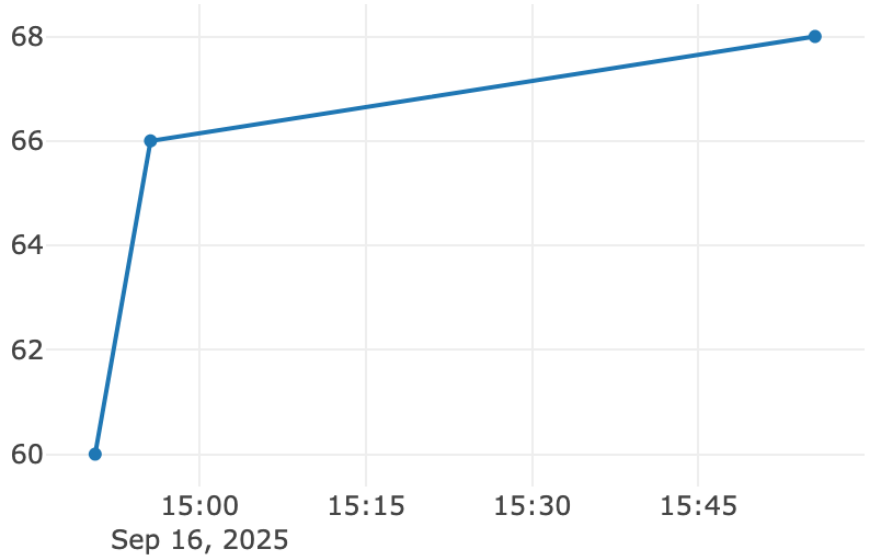
plotly



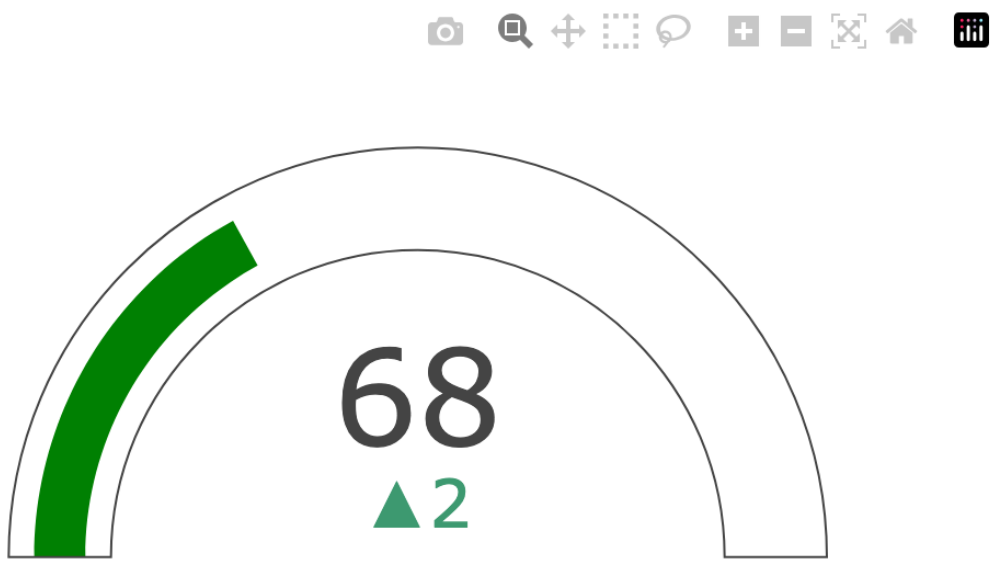
JS

JavaScript

This is the expected end result for the web app



Graph showing the heart rate reading throughout the current day



Indicator graph showing the last reported heart rate. As compared to previos value. Min/Max should be human min/max HR.

Useful Links

JavaScript, JSON, REST API:

- https://www.w3schools.com/js/default.asp
- https://www.youtube.com/watch?v=6OhMbf2v_jI

Plotly:

- https://www.youtube.com/watch?v=GVrjv9ajsJ0
- https://plotly.com/javascript/
- https://plotly.com/javascript/indicator/
- https://plotly.com/javascript/line-charts/

And Many More

We use FHIR Release 4, so when reading the standards, make sure to read the R4 version!

Useful Links, cont.

Data Modelling:

- https://www.youtube.com/watch?v=wOD02sezmX8
- https://www.youtube.com/watch?v=n3mHfQft5P8
- https://www.youtube.com/watch?v=uPOGPL2C0_8

FHIR

- https://hl7.org/fhir/R4/ - The official standard R4
- https://www.youtube.com/watch?v=OIt0GrCPu8k
- https://www.youtube.com/watch?v=YbQcJj1GqH0
- https://www.youtube.com/playlist?list=PLUr-PTsPYKV4SHhszovqJ3v4hEhRezqzo (first three is enough)

Understanding `Promise` in Javascript and =>

```
<!DOCTYPE html>
<html>

<head>
<title>Test</title>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
<script src="jqFhir.js"></script>
</head>

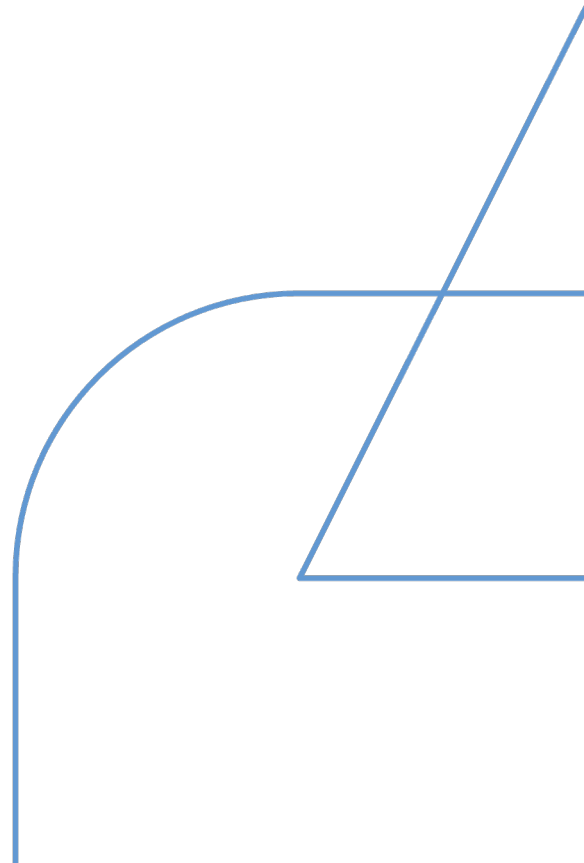
<body>
<h1>Test</h1>
Watch this script in the console log in your browser!

<script>
	console.log("First!");

	var client = jqFhir({ baseUrl: 'https://hapim.app.cloud.cbh.kth.se/fhir' });

	client.read({
		id: 'Patient/12'
	}).then(p => {
		console.log("Second!");
	});

	console.log("Third!");
</script>
</body>
</html>
```



Our FHIR Database

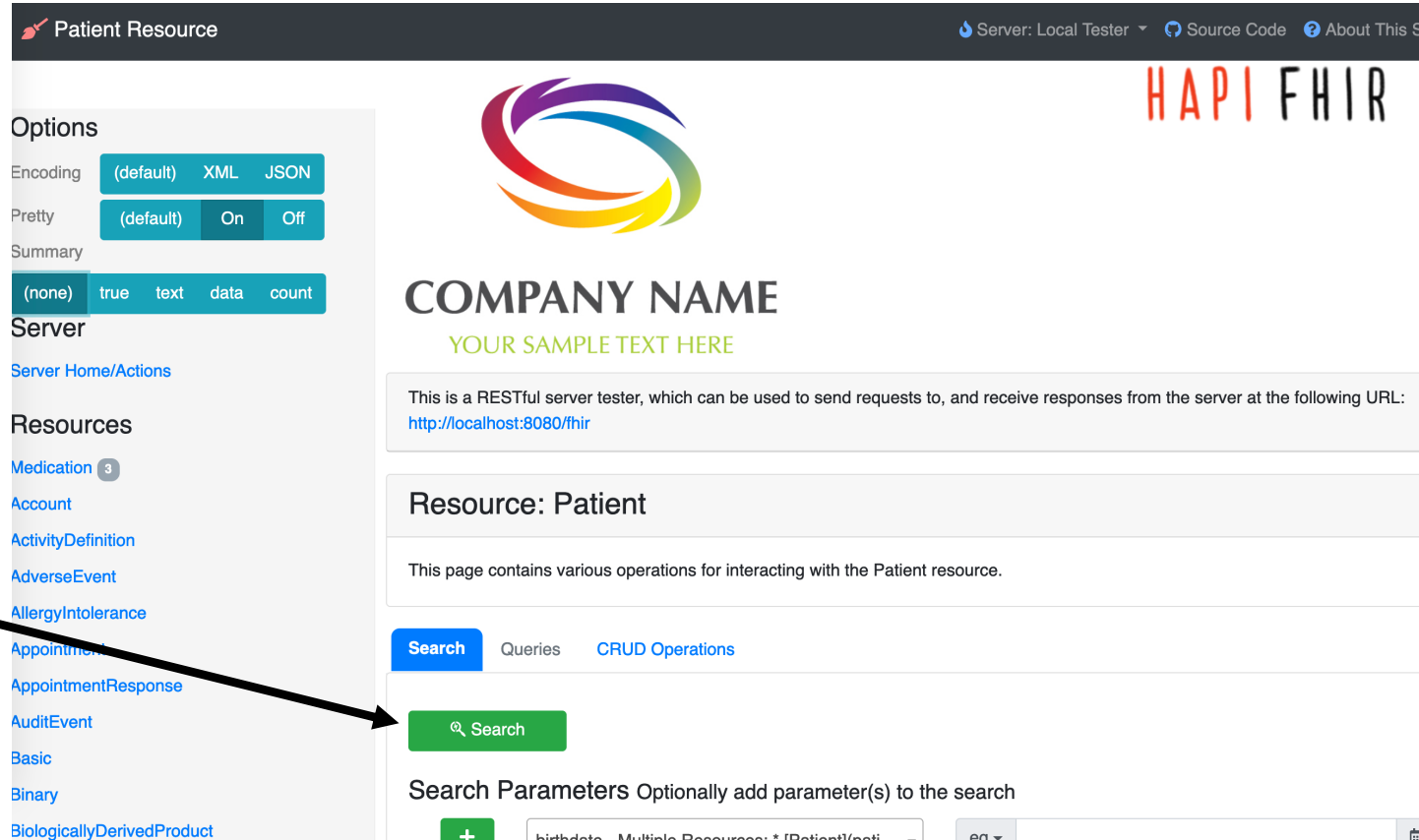


https://hapifhir.io/

- BASE URL: https://hapim.app.cloud.cbh.kth.se/fhir
- https://hapim.app.cloud.cbh.kth.se/
- FHIR Release R4

Click ’Patient’
Then click Search.

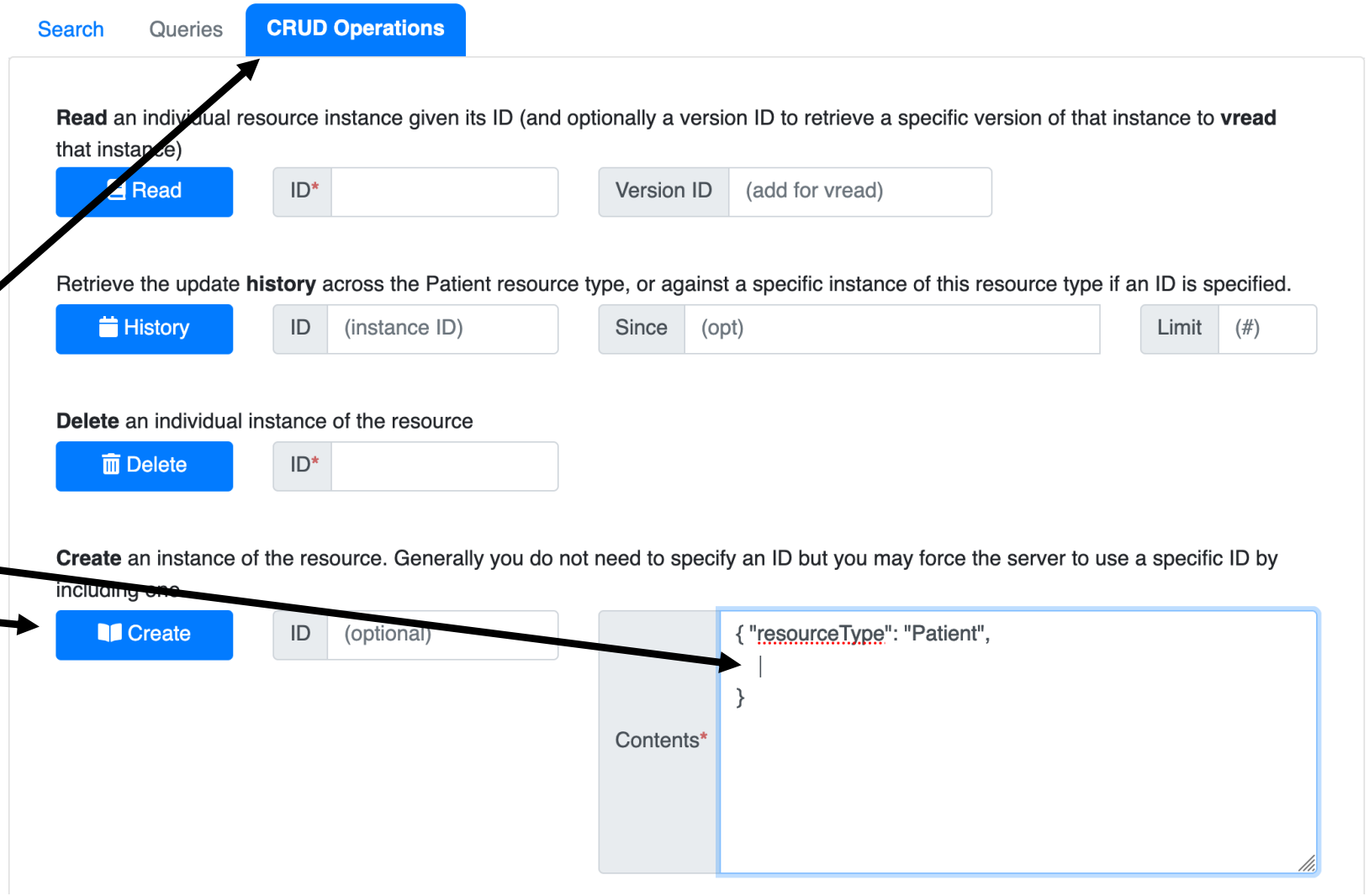
You should see all patients in the DB



Create a Patient

Click ’Patient’
Click ’CRUD Operations’
Fill in Contents
Click ’Create’

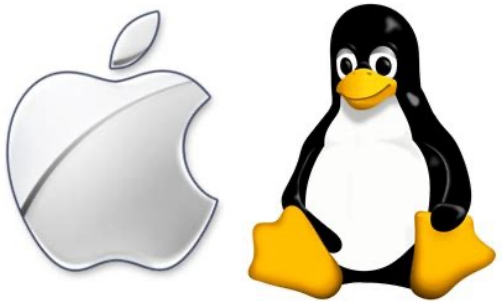
A Patient will be added if successful!



Command Line version of creating a Resouce

```
BASEURL="https://hapim.app.cloud.cbh.kth.se/fhir"

curl -X 'POST' \
  "${BASEURL}/Patient" \
  -H 'accept: application/fhir+json' \
  -H 'Content-Type: application/fhir+json' \
  -d '{
  "resourceType": "Patient",
  "active": true,
  "name": [
    {
      "use": "official",
      "family": "Mune",
      "given": [
        "Peterson"
      ],
      "prefix": [
        "Mr"
      ]
    }
  ],
  "telecom": [
    {
      "system": "phone",
      "value": "584785"
    }
  ],
  "gender": "male",
  "birthDate": "2022-01-02",
  "address": [
    {
      "city": "EMBU",
      "country": "KENYA"
    }
  ]
}'
```



This example shows how it is done in Linux.

The Task

Only work on your own Patient!
We all share the same DB

- Create your own Patient Resource in the KTH HAPI FHIR database.
  - Create a JSON-representation and upload it using HAPI developer UI (or curl or Insomnia).
  - Which attributes do we need to fill? Which ones can we leave out? Why?
- Verify that you can see this with `patientview.html` and `observationview.html`
  - `file:///<YourPath>/patientview.html?id=<YourPatientID>`

The Task, cont.

Only work on your own Patient!
We all share the same DB

- Update the `BLEGW.html` file to create an Observation Resource and send it with JavaScript code to the KTH HAPI FHIR DB.
  - Which attributes do we need to fill? Which ones can we leave out? Why?
  - How do we fill the Observation `code`? And `method`? Find the right values? Not only by looking at other's examples. Find them in the standards!
  - Send heart rate data once every 1 minute, and when we click the button "Send HR Value".
  - Just send the last reading from the Micro:Bit.
- Update `graph.html` to download today's heart rate observations and create a curve.
  - Show the latest value in the right part.
  - Load new values once every 10 seconds and display in the graph (i.e., live updates)
  - Make sure it displays today's reading according to local time (time zone/day light saving time, etc).
  - Make sure only HR values are displayed.

Getting started

- Download the ZIP-file from KTH Canvas, it contains the files you need to get started.
- We build using this JS-library:
  - https://github.com/FHIR/fhir.js
  - You do not need to install it. Just use the file jqFhir.js, which is part of the ZIP-file
- A partial front-end is created, where you can view patients and observations. You may look at the source code to get inspiration!
- Use the HAPI developer front-end to verify what is in the DB:
  - https://hapim.app.cloud.cbh.kth.se/